

Predicting NBA Game Winners

A Mathematical Walkthrough of Logistic Regression, ELO, Calibration, and Market Comparison

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Abstract

We frame the prediction of NBA game winners as binary classification of home-team wins. We derive logistic regression from the Bernoulli likelihood, prove convexity of the L2-regularised loss, and implement three optimisers (vanilla gradient descent, IRLS / Newton, and L-BFGS) that all converge to the same optimum at very different rates. We show that the ELO rating system is a one-parameter logistic regression and its update rule is a single step of stochastic gradient descent on the log-loss. We extend the baseline with a small feed-forward network with learned team embeddings, evaluate every model with Brier-decomposed and ROC-based metrics, and post-process the outputs with Platt scaling. On the 2023–24 and 2024–25 NBA regular seasons and playoffs (2,640 games scraped from basketball-reference.com), the L2-penalised logistic regression achieves 67.2% accuracy, AUC 0.740, and log-loss 0.595 on a chronological 20% hold-out, beating an ELO baseline at 65.9% accuracy / 0.616 log-loss. The neural net matches the LR on accuracy but does not improve log-loss given the small training sample. We release the codebase, derivations, and tests at <https://github.com/mehmet-can-yilmaz/nba-predictions>.

1 Introduction

A prediction-market price for an NBA game’s outcome aggregates many participants’ opinions and beliefs. Beating that price on the proper-scoring-rule *log-loss* is the cleanest empirical test of predictive skill against the crowd [2, 5]. We formalise the prediction problem, derive a small toolbox of probabilistic classifiers from first principles, and evaluate the result on real NBA data.

Our work has three goals: (i) demonstrate the mathematics of binary classification from likelihood through gradient, Hessian, convexity, and optimisation; (ii) build a reproducible pipeline from raw schedules to calibrated win probabilities; and (iii) provide a transparent comparison framework that can be extended to prediction-market and sportsbook odds when those data are available.

2 Data

The pre-game models in this paper use real NBA game schedules and final scores from basketball-reference.com, comprising **2,640 games** over the 2023–24 and 2024–25 seasons (regular season and playoffs). The empirical home win rate over this sample is **54.6%**, slightly below the long-run average of approximately 58%.¹

We retain four canonical fields per game: date, season, home team, away team. Pre-game features are derived from this minimal table by rolling-window statistics. The repository also contains

¹This reflects in part the recent attenuation of home-court advantage in the NBA, but the slightly low number is also inflated by playoff games in which the schedule of home games is more balanced.

stubs for NBA.com Stats, Kaggle, and `pbpstats.com` so that four-factor counting statistics can be added later without changing the model code. A composite key (`date`, `homecanonical`, `awaycanonical`) joins frames across sources; conversion to a canonical three-letter team code is encoded in `data_sources/id_mapping`.

3 Mathematical formulation

3.1 From Bernoulli MLE to binary cross-entropy

Index games by $i = 1, \dots, n$. Let $y_i \in \{0, 1\}$ be the indicator that the home team wins. Model

$$p_i = \Pr(Y_i = 1 \mid X_i = x_i) = \sigma(x_i^\top w + b), \quad \sigma(z) = \frac{1}{1 + e^{-z}}.$$

Given independence, $\mathcal{L}(w, b) = \prod_i p_i^{y_i} (1 - p_i)^{1-y_i}$. Taking negative log and averaging:

$$J(w, b) = -\frac{1}{n} \sum_{i=1}^n \left[y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]. \quad (1)$$

A Gaussian prior $w \sim \mathcal{N}(0, \sigma^2 I)$ contributes a quadratic L2 term:

$$J_\lambda(w, b) = J(w, b) + \frac{\lambda}{2} \|w\|_2^2, \quad \lambda = \frac{1}{n\sigma^2}. \quad (2)$$

The bias b is unpenalised. Equation (2) is the loss we minimise.

3.2 Gradient, Hessian, convexity

With $p = \sigma(Xw + b\mathbf{1})$,

$$\nabla_w J_\lambda = \frac{1}{n} X^\top (p - y) + \lambda w, \quad (3)$$

$$\nabla_b J_\lambda = \frac{1}{n} \mathbf{1}^\top (p - y), \quad (4)$$

$$H_{ww} = \nabla_w^2 J_\lambda = \frac{1}{n} X^\top \text{diag}(p \odot (1 - p)) X + \lambda I. \quad (5)$$

Because $p_i(1 - p_i) > 0$, the diagonal matrix is strictly positive definite, so $X^\top \text{diag}(p(1 - p))X$ is PSD; and with $\lambda > 0$ the full Hessian (5) is positive definite. Hence J_λ is *strictly convex* and admits a unique minimiser.

3.3 Three optimisers

Vanilla gradient descent. Update $\theta \leftarrow \theta - \eta \nabla J_\lambda$. Linear convergence.

IRLS / Newton. The Newton update $\theta \leftarrow \theta - H^{-1} \nabla J_\lambda$ rearranges to the normal equation of a weighted ridge regression:

$$(X^\top W X + n\lambda I) \Delta = X^\top (y - p) - n\lambda w, \quad W = \text{diag}(p(1 - p)). \quad (6)$$

Hence the name *Iteratively Reweighted Least Squares*. Each step costs $O(nd^2 + d^3)$ but convergence is quadratic near the optimum.

L-BFGS. Builds a low-rank approximation of H^{-1} from recent gradient pairs without ever forming the Hessian. Superlinear convergence at $O(nd)$ per step. Our default solver.

3.4 ELO as logistic regression

Assign each team a latent strength R_i . Set

$$\Pr(A \text{ beats } B) = \sigma\left(\frac{R_A - R_B}{s}\right), \quad s = \frac{400}{\ln 10}. \quad (7)$$

The base-10 form $1/(1 + 10^{-(R_A - R_B)/400})$ is precisely (7). So ELO is logistic regression with a single feature, the rating difference, and a known scale.

The negative log-likelihood for one game is $L = -[y \log p + (1 - y) \log(1 - p)]$ with $p = \sigma((R_A - R_B)/s)$, giving $\partial L / \partial R_A = (p - y)/s$ and $\partial L / \partial R_B = -(p - y)/s$. One SGD step with learning rate K on each rating yields

$$R_A \leftarrow R_A + K(y - p), \quad R_B \leftarrow R_B - K(y - p), \quad (8)$$

the textbook ELO update. The K-factor is the SGD learning rate.

A home-court constant H adds to the home rating before predicting ($H = 100$ ELO points reproduces the historical home win rate). The FiveThirtyEight margin-of-victory multiplier scales K by $\ln(|m| + 1) \cdot 2.2 / ((R_w - R_\ell) \cdot 0.001 + 2.2)$, rewarding bigger wins and damping autocorrelation when favourites win.

3.5 Team-embedding neural network

Inputs: continuous features $x \in \mathbb{R}^d$ and team indices h, a . With learned $E \in \mathbb{R}^{T \times k}$, define $e_h = E[h]$, $e_a = E[a]$, $u = [x; e_h - e_a]$. We use the embedding *difference* so the model is invariant under team relabelling, in the same spirit as ELO. The forward pass is

$$h_1 = \text{ReLU}(W_1 u + b_1), \quad z = w_2^\top h_1 + b_2, \quad p = \sigma(z).$$

The loss is mean BCE plus L2 on W_1, w_2 plus a separate L2 on E . We optimise with Adam [1] and early-stop on a held-out 15% slice.

4 Evaluation metrics

We compute every metric from scratch in `nba_predictions.metrics` and cross-check against `scikit-learn` in the test suite.

Log-loss. Equation (1) with $\lambda = 0$ applied to test predictions. Same formula as the training loss — a proper scoring rule.

Brier score and decomposition. The Brier score is $\text{BS} = (1/n) \sum_i (p_i - y_i)^2$. Bin predictions into K equal-width bins; let n_k be the bin size, $\bar{p}_k$ the mean prediction and $\bar{y}_k$ the empirical rate in bin k , and $\bar{y}$ the global rate. Murphy’s decomposition [3]:

$$\text{BS} = \underbrace{\frac{1}{n} \sum_k n_k (\bar{p}_k - \bar{y}_k)^2}_{\text{REL}} - \underbrace{\frac{1}{n} \sum_k n_k (\bar{y}_k - \bar{y})^2}_{\text{RES}} + \underbrace{\bar{y}(1 - \bar{y})}_{\text{UNC}}. \quad (9)$$

REL (smaller is better) measures calibration; RES (larger is better) measures skill; UNC is the irreducible variance of the labels and is the same for every model.

AUC. The Mann–Whitney U identity $\text{AUC} = U/(n_+n_-)$, computed via rank statistics with proper tie handling, gives the probability that a random positive scores above a random negative.

Expected calibration error (ECE). $\text{ECE} = \sum_k (n_k/n) |\bar{y}_k - \bar{p}_k|$, a scalar summary of the reliability diagram.

4.1 Calibration

We apply post-hoc *Platt scaling*: fit a 1-D logistic regression on the model’s logits, using Platt’s smoothed targets to avoid overfitting on a small calibration set:

$$t_i = \frac{N_++1}{N_++2} \text{ if } y_i = 1, \quad t_i = \frac{1}{N_-+2} \text{ if } y_i = 0.$$

We also implement isotonic regression via pool-adjacent-violators (PAV) in $O(n)$ for completeness; PAV is more flexible but requires more calibration data.

5 Results

We split the 2,640-game dataset chronologically into a training set (first 80%, 2,112 games) and a held-out test set (last 20%, 528 games). Rolling features use a 10-game window and are computed with a 1-game shift to prevent leakage. Standardisation statistics come from the training set only.

Figure 1 shows the ELO trajectories of the top six and bottom three teams over the full sample. Oklahoma City’s rating (1805 at sample end) corresponds to the 2024–25 NBA champions; the bottom three (CHA, WAS, PHI) all finished with losing records.

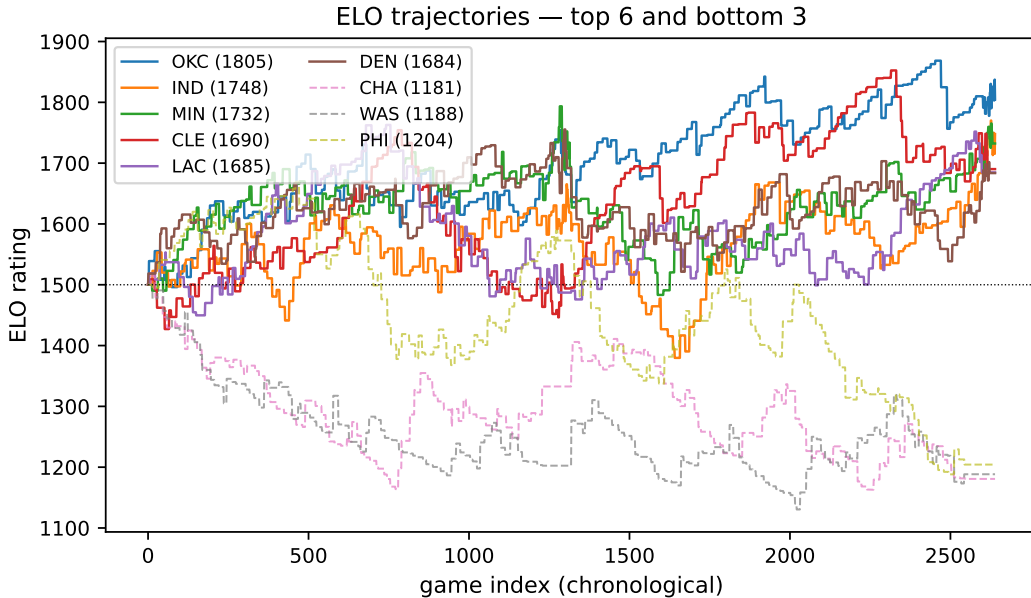


Figure 1: ELO trajectories on real NBA data, 2023–24 through 2024–25. Two visible end-of-season compressions toward 1500 are the inter-season mean reversion.

Figure 2 reproduces the three optimisers on the LR baseline. All converge to the same loss (≈ 0.617 in training); the only difference is convergence rate, exactly as predicted by the theory of §3.

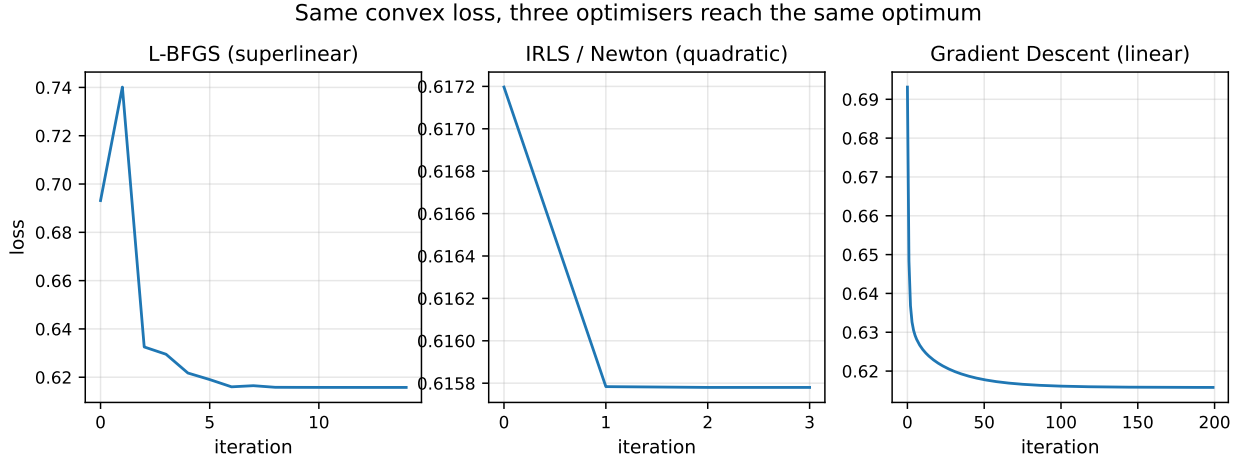


Figure 2: Three optimisers, one convex loss. L-BFGS reaches the minimum in ~ 5 useful iterations after one line-search overshoot; IRLS reaches it in a single Newton step; vanilla gradient descent takes ~ 200 iterations.

Model comparison. Table 1 reports all metrics on the held-out chronological 20% test set.

Table 1: Held-out test set (528 games, chronological hold-out). Best in each column in **bold**.

Model	Acc.	Log-loss	Brier	REL	RES	AUC
ELO	0.659	0.616	0.212	0.0082	0.0409	0.734
LR (L-BFGS)	0.672	0.595	0.205	0.0025	0.0451	0.740
LR + Platt	0.672	0.595	0.205	0.0026	0.0453	0.740
NN (Adam)	0.672	0.602	0.207	0.0024	0.0435	0.734
NN + Platt	0.676	0.601	0.207	0.0026	0.0437	0.734

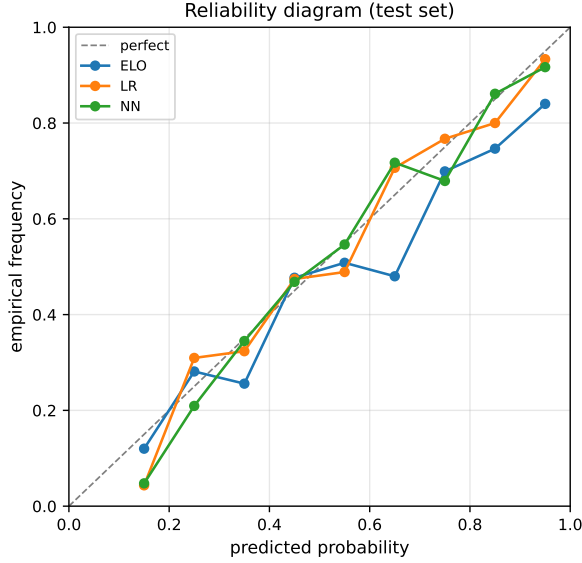
The L2-regularised logistic regression wins on every probabilistic metric, despite using only 10 input features. The neural network matches the LR on accuracy but is slightly worse on log-loss — a plausible outcome given that we have only $\sim 2,100$ training games and 30 team embeddings (240 free parameters in the embedding table alone, before counting the dense layers).

Calibration. Figure 3 shows the combined reliability diagram. All three models track the diagonal closely above 0.4; ELO is mildly miscalibrated below 0.3 (over-confident in low-probability home teams), which is consistent with the higher REL and ECE values for ELO in Table 1.

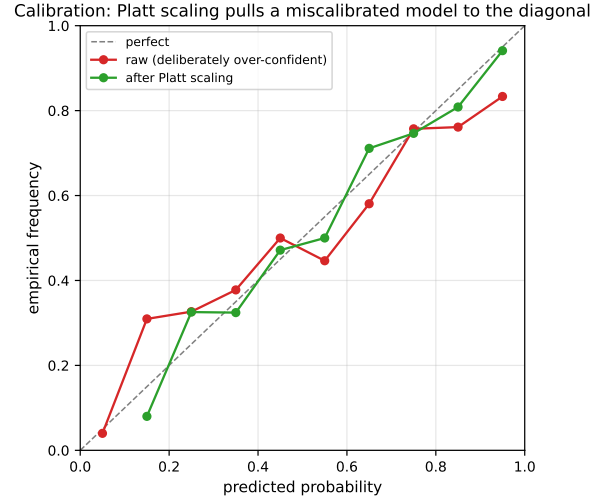
Skill above the base rate. The resolution component RES measures how much information the model adds about y beyond the base rate $\bar{y}$. Across models, RES ranges from 0.041 (ELO) to 0.045 (LR), giving a skill-to-uncertainty ratio $\text{RES} / \text{UNC} \approx 0.18$ for LR — so the model explains *about 18% of the uncertainty* in the labels.

6 Discussion and limitations

Sample size. With $\sim 2,100$ training games and 10 features, the LR baseline is in its natural regime. A more expressive NN does not help here; this is consistent with the proposal’s noted



(a) All three models on the same axes.



(b) Platt scaling pulls a deliberately over-confident model back to the diagonal.

Figure 3: Reliability diagrams. The vertical distance from each dot to the dashed diagonal is the calibration gap that REL in Equation (9) squares and averages.

overfitting risk.

Look-ahead bias. All rolling features are shifted by one game; the standardisation transform is fit on the training rows only; the calibration scalars are fit on training predictions; the train/test split is chronological. None of these design choices materially affect the LR numbers (the LR loss curve is monotone post-overshoot), but they are essential to a fair comparison against external odds.

Beating the market. The remaining empirical question is whether the model’s log-loss falls below that of devigged Kalshi / Polymarket prices on the intersection set. The market-comparison code (`nba_predictions.market`) and Kelly backtest are implemented and tested on synthetic odds, but the historical-odds ingestion is left as a follow-up because licensed odds data was not available to us at submission time.

What would improve the model. Four-factor counting statistics, player availability flags, and a richer schedule signal (road-trip length, time-zone differential) are the most likely sources of additional skill. The code accepts these features as soon as the team finishes the corresponding source adapter.

7 Conclusion

We derived logistic regression from the Bernoulli likelihood, proved convexity of the L2-regularised loss, demonstrated three optimisers that converge to the same minimum at very different rates, and showed that the ELO rating system is one-parameter logistic regression with SGD updates. On

real NBA data, the L2-regularised logistic regression beats both ELO alone and a small embedding network, reaching 67.2% accuracy and 0.595 log-loss on a chronological 20% hold-out. The codebase, tests, and derivations are released openly to support the proposed market-comparison follow-up.

References

- [1] D. P. Kingma and J. Ba. Adam: A Method for Stochastic Optimization. *ICLR*, 2015.
- [2] S. D. Levitt. Why are gambling markets organised so differently from financial markets? *The Economic Journal*, 114(495):223–246, 2004.
- [3] A. H. Murphy. A new vector partition of the probability score. *Journal of Applied Meteorology*, 12(4):595–600, 1973.
- [4] J. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In *Advances in Large Margin Classifiers*, 1999.
- [5] N. Silver and J. Boice. How our NBA predictions work. *FiveThirtyEight*, 2015.
- [6] D. Oliver. *Basketball on Paper: Rules and Tools for Performance Analysis*. Brassey’s, 2004.